Experiment No.	1
Experiment Title	Iris Flower Classification using K-Nearest Neighbours
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1 Objective

The aim of this experiment is to classify iris flowers into three species—*Iris setosa*, *Iris versicolor*, and *Iris virginica*—using the K-Nearest Neighbours (KNN) algorithm. The specific objectives are to perform exploratory data analysis on the Iris dataset, select the most informative features using Chi-Square and ANOVA tests, train a KNN classifier on the selected features, and evaluate its performance using standard metrics.

2 Problem Statement

Given four morphological measurements of iris flowers (sepal length, sepal width, petal length, and petal width), predict which of the three species a flower belongs to. The target column carries integer class labels (0 = setosa, 1 = versicolor, 2 = virginica). Because class labels are provided in the dataset, this is a **supervised multi-class classification** problem.

3 Dataset Description

Dataset Name	Iris Dataset (CSV)
Source	UCI Machine Learning Repository / Kaggle
Number of Samples	150
Number of Features	4 numeric + 1 target (species)
Number of Classes	3 (setosa, versicolor, virginica)
Class Balance	Perfectly balanced — 50 samples per class
Missing Values	None
Duplicate Rows	3
Train-Test Split	80 % / 20 % (stratified, random_state=42)

Features:

- `sepal_length` (cm) — length of the sepal leaf
- `sepal_width` (cm) — width of the sepal leaf
- `petal_length` (cm) — length of the petal leaf
- `petal_width` (cm) — width of the petal leaf

4 Exploratory Data Analysis

Statistical Summary

Statistic	Sepal Length	Sepal Width	Petal Length	Petal Width
Mean	5.843	3.057	3.758	1.199
Std Dev	0.828	0.436	1.765	0.762
Min	4.300	2.000	1.000	0.100
Max	7.900	4.400	6.900	2.500

Table 2: Descriptive statistics of the four numeric features

Petal length and petal width have much higher standard deviations than sepal measurements, suggesting greater variability and therefore stronger discriminative power.

Feature Distributions

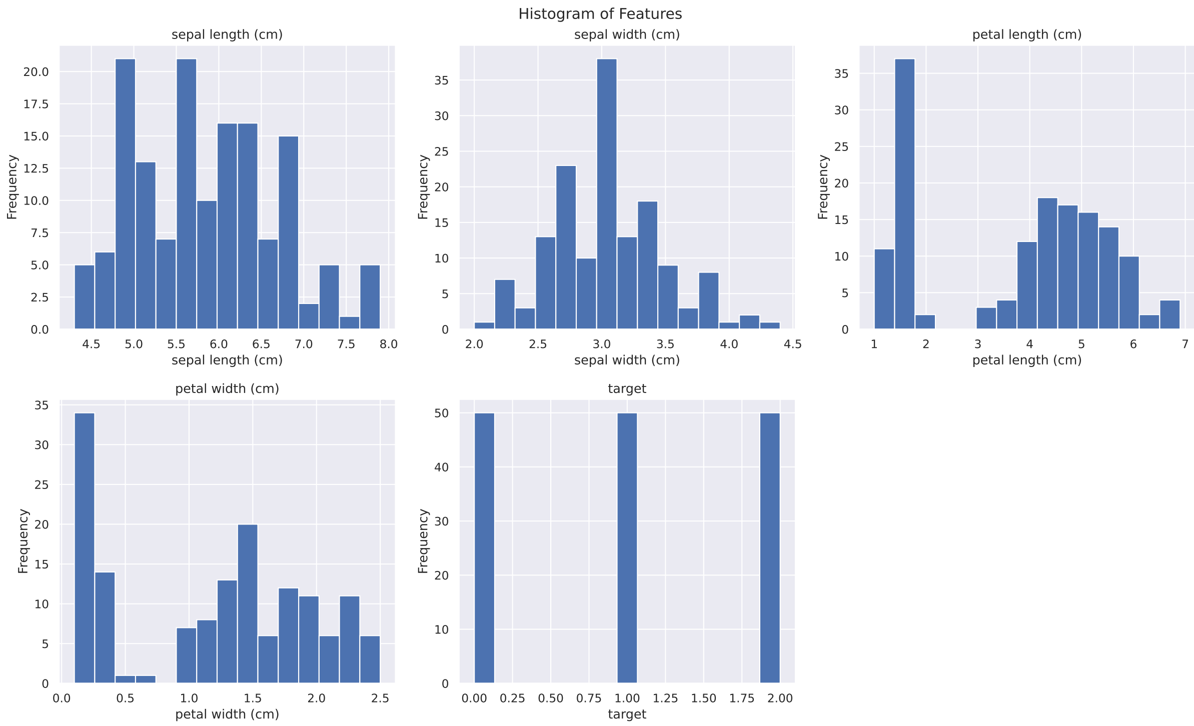

Figure 1: Histograms of all numeric features and the target column

Inference: Sepal length follows a rough bell curve, while petal length and petal width show clear bimodal distributions that correspond to the setosa class (very small petals) being separated from the other two species. This bimodality is a strong visual hint that petal features will dominate classification.

Class Distribution

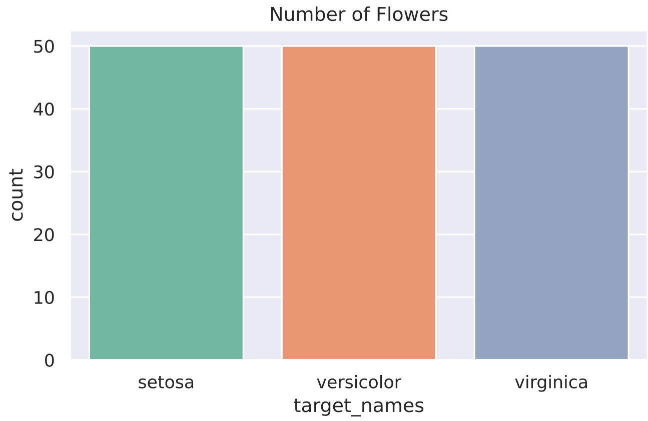

Figure 2: Count of samples per species

Inference: The dataset is perfectly balanced with exactly 50 samples per class. There is no class imbalance problem, so accuracy is a fair performance metric and no resampling is needed.

Feature vs Target Scatter (Relational Plots)

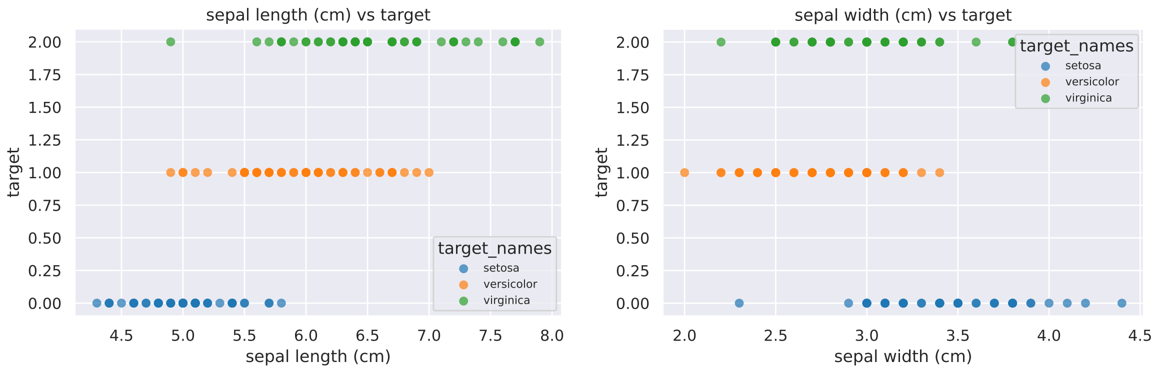

Figure 3: Sepal length and sepal width vs target class

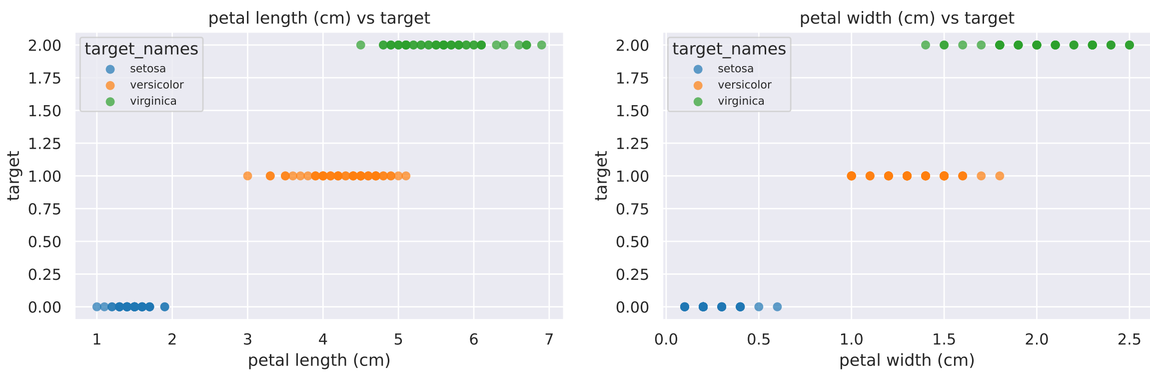

Figure 4: Petal length and petal width vs target class

Inference: Setosa is perfectly separated from the other two classes in the petal length and petal width plots, confirming these are the most discriminative features. Sepal width shows considerable overlap between all three classes and is likely the weakest predictor.

Pair Plot

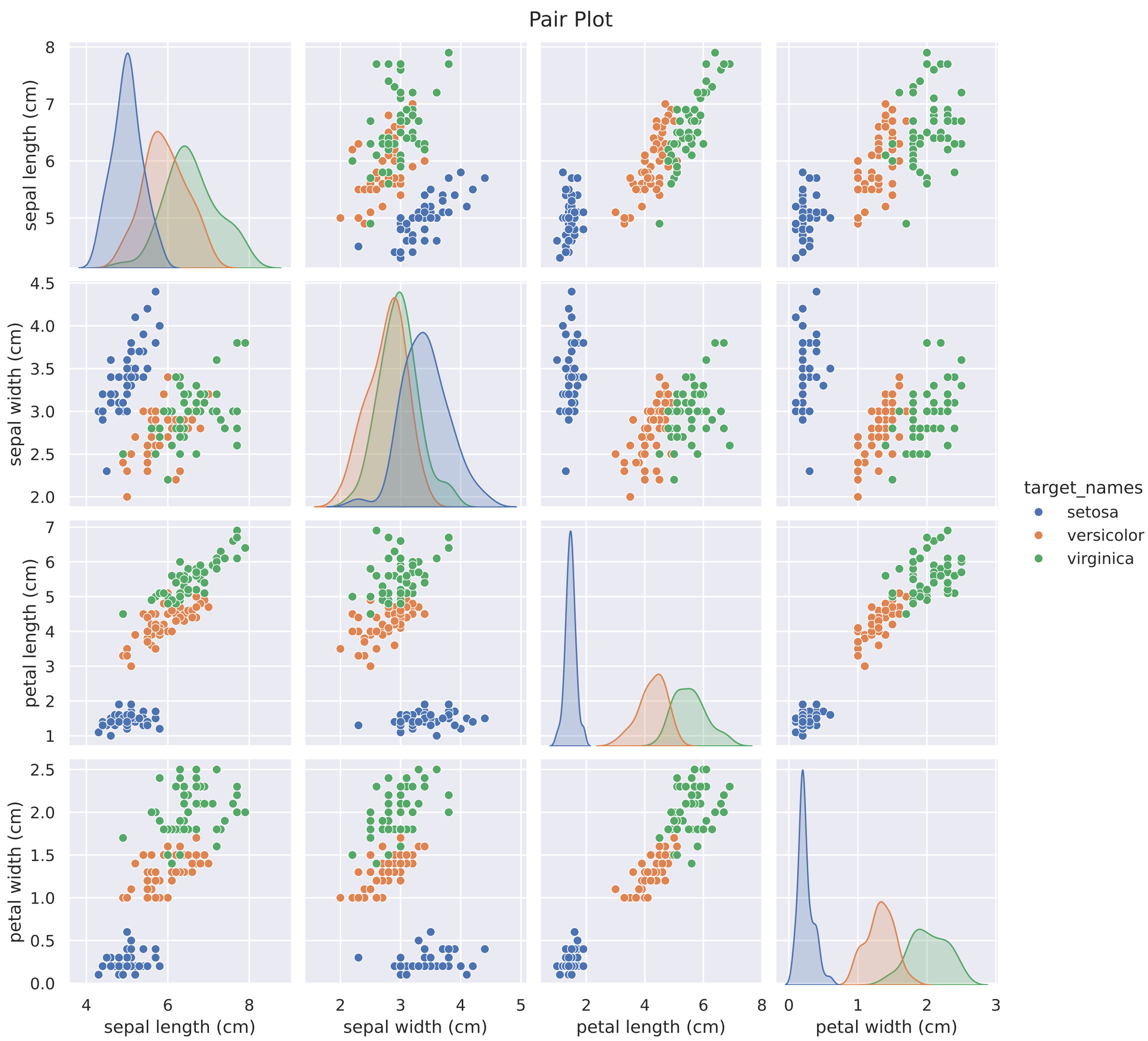

Figure 5: Pair plot of all numeric features coloured by species

Inference: The pair plot confirms that setosa is linearly separable from the other two species in almost every feature combination. Versicolor and virginica overlap slightly, particularly in sepal-based axes, but are well separated on petal measurements. The diagonal KDE plots show distinct peaks per species for petal features.

Box Plot

Figure 6: Petal length distribution per species

Inference: Setosa has a very compact petal length distribution (1.0–1.9 cm), while virginica spans the highest range (∼4.5–6.9 cm). Versicolor lies in between. The non-overlapping interquartile ranges confirm petal length alone is a strong class separator.

Scatter Plot — Petal Length vs Petal Width

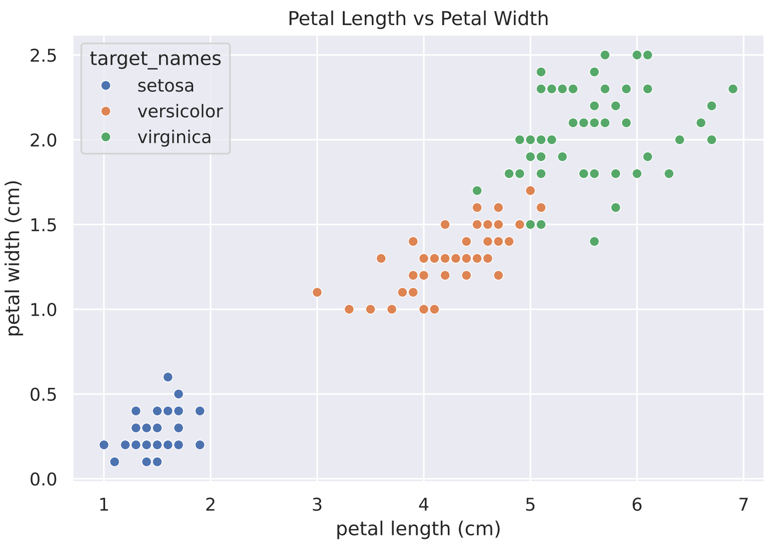

Figure 7: Petal length vs petal width coloured by species

Inference: Setosa occupies the bottom-left corner, well isolated from the other two species. Versicolor and virginica overlap marginally at their boundary but are otherwise separable. This two-feature scatter plot supports selecting petal length and petal width as the final feature pair for modelling.

Outlier Detection

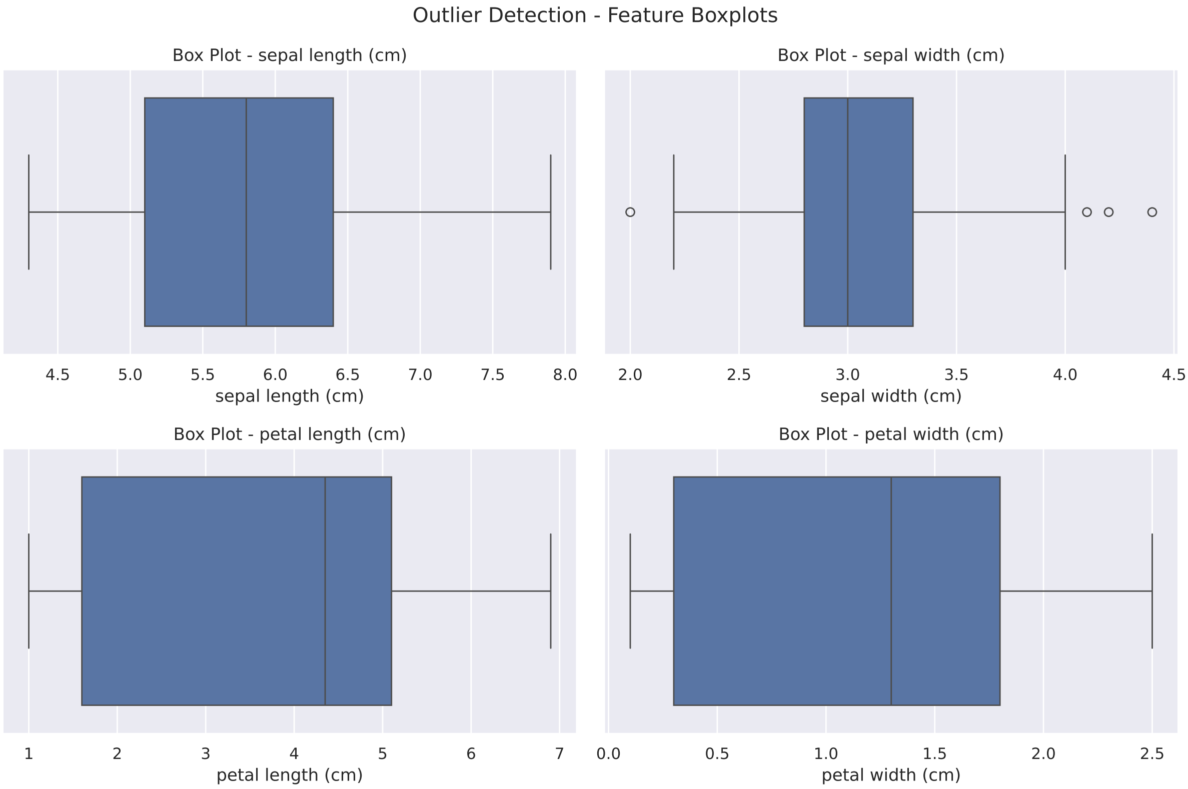

Figure 8: Box plots for individual feature outlier detection

Inference: Sepal width shows a few outliers beyond the upper whisker, particularly at values above 4.0 cm. Petal features have minimal outliers. Since the dataset is small (150 samples) and the outliers appear valid (natural measurement variation), they are retained without removal.

Correlation Heatmap

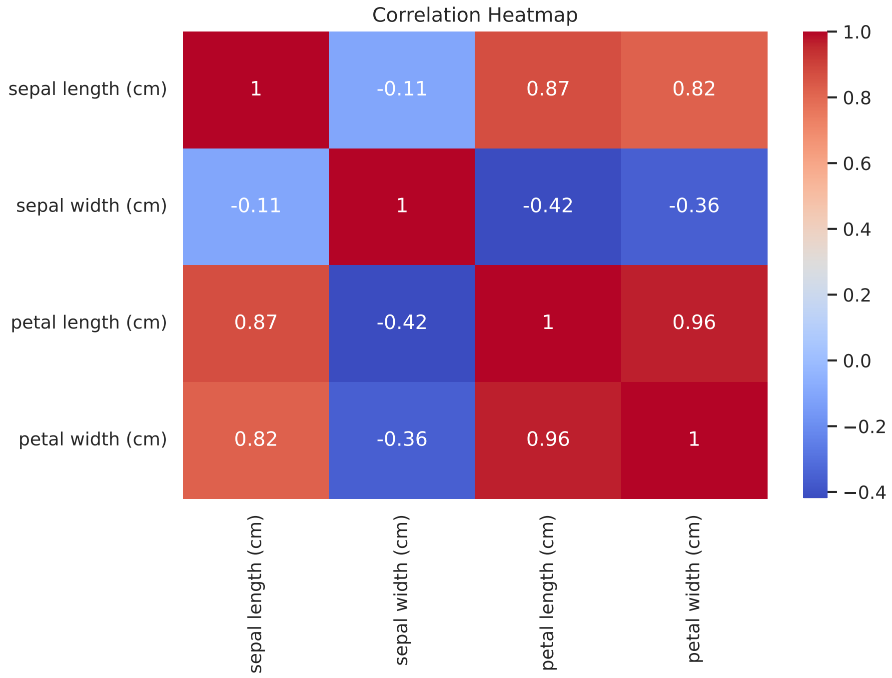

Figure 9: Pearson correlation heatmap (coolwarm palette)

Inference: Petal length and petal width are highly correlated (0.96), which makes sense anatomically. Both petal features are also moderately correlated with sepal length (0.87 and 0.82 respectively). Sepal width is weakly or negatively correlated with the other features, further confirming its limited discriminative value.

5 Data Preprocessing

1. **Missing value check:** No missing values exist in the dataset; no imputation was needed.
2. **Label encoding:** The `species` column was mapped to integer targets: setosa → 0, versicolor → 1, virginica → 2.
3. **Feature scaling:** `StandardScaler` was applied to all four features to give zero mean and unit variance. Standardisation is important for KNN because it is a distance-based algorithm; without it, features measured in different ranges (e.g., petal length 1–7 vs petal width 0.1–2.5) would contribute unequally to the distance calculation.
4. **Feature selection:** SelectKBest with Chi-Square (on raw values) and ANOVA F-test (on scaled values) were used to rank features and select the top two.
5. **Train-test split:** 80/20 stratified split (random_state = 42), giving 120 training samples and 30 test samples with equal class proportions in each split.

6 Feature Selection

Two statistical tests were applied to rank the four features:

Chi-Square Test measures dependence between each feature and the class label using raw (non-negative) feature values. A higher score means the feature is more informative.

ANOVA F-Test measures how much the feature mean differs across classes relative to within-class variance. Applied on standardised features.

Feature	Chi-Square Score	ANOVA Score
sepal length (cm)	10.82	119.26
sepal width (cm)	3.59	47.36
petal length (cm)	**116.17**	**1179.03**
petal width (cm)	**67.24**	**959.32**

Table 3: Feature importance scores from Chi-Square and ANOVA tests

Both tests consistently rank **petal length** and **petal width** as the top two features by a large margin. These two features were selected for model training using `SelectKBest(k=2)`.

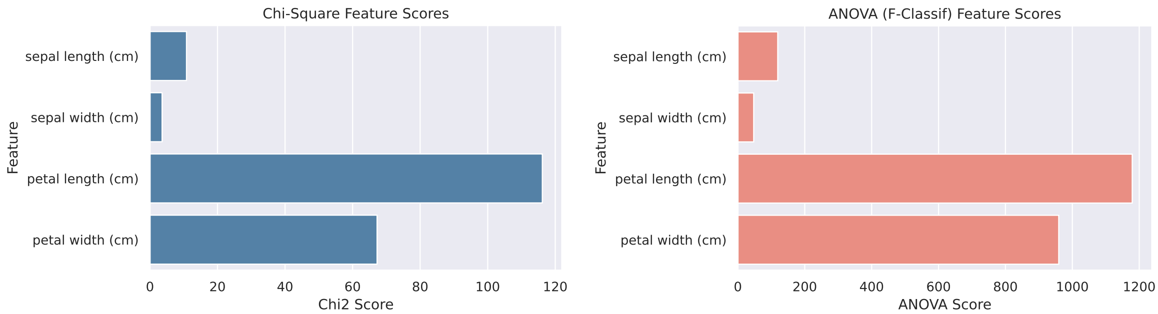

Figure 10: Chi-Square scores (left) and ANOVA F-scores (right) for all features

Inference: Petal length has the highest score in both tests by a significant margin. Sepal width has the lowest scores in both, confirming it is the least useful feature for distinguishing species. The agreement between two independent tests strengthens confidence in the selected feature pair.

7 Machine Learning Algorithm

7.1 K-Nearest Neighbours (KNN)

KNN is a non-parametric, instance-based (lazy) learning algorithm. It stores the entire training set and classifies a new sample by finding the k most similar training points under a chosen distance metric, then assigns the majority class label:

$$\hat{y} = \text{mode}\{y_i : x_i \in \mathcal{N}_k(x)\}$$

The default distance metric used by scikit-learn's `KNeighborsClassifier` is the Minkowski distance, which reduces to Euclidean distance when $p = 2$:

$$d(x, x') = \left(\sum_{j=1}^{d} |x_j - x'_j|^2 \right)^{1/2}$$

Default hyperparameters used:

- $k = 5$ (number of neighbours)
- Weight function: `uniform` (all neighbours contribute equally)
- Algorithm: `auto` (selects KDTree, BallTree, or brute force automatically)
- Distance metric: Minkowski ($p = 2$, i.e., Euclidean)

Advantages:

- No explicit training phase — fast to set up.
- Naturally handles multi-class problems.
- Makes no assumptions about the underlying data distribution.

Limitations:

- Prediction time is $O(nd)$ for brute force, making it slow on large datasets.
- Sensitive to irrelevant or unscaled features.
- Memory-intensive: must store all training samples.

8 Experimental Setup

Python Version	3.12
Scikit-learn Version	1.8.0
IDE	Jupyter Notebook
Operating System	Ubuntu 24.04
Dataset	iris.csv (150 samples)

Hyperparameter Configuration

Parameter	Value
n_neighbors	5 (default)
weights	uniform
algorithm	auto
metric	minkowski ($p = 2$, Euclidean)
Selected Features	petal length (cm), petal width (cm)
Scaler	StandardScaler

Table 4: KNN hyperparameter configuration

9 Results

Performance Metrics

Class	Precision	Recall	F1-Score
setosa	1.00	1.00	1.00
versicolor	1.00	0.90	0.95
virginica	0.91	1.00	0.95
Accuracy	0.97 (29/30 correct)		
Macro Avg	0.97	0.97	0.97

Table 5: Classification report on the test set

The KNN classifier achieves **97% accuracy** using only the top 2 petal features. Setosa is perfectly classified; the single error is one versicolor sample misclassified as virginica.

10 Result Visualization

1. Correlation Matrix

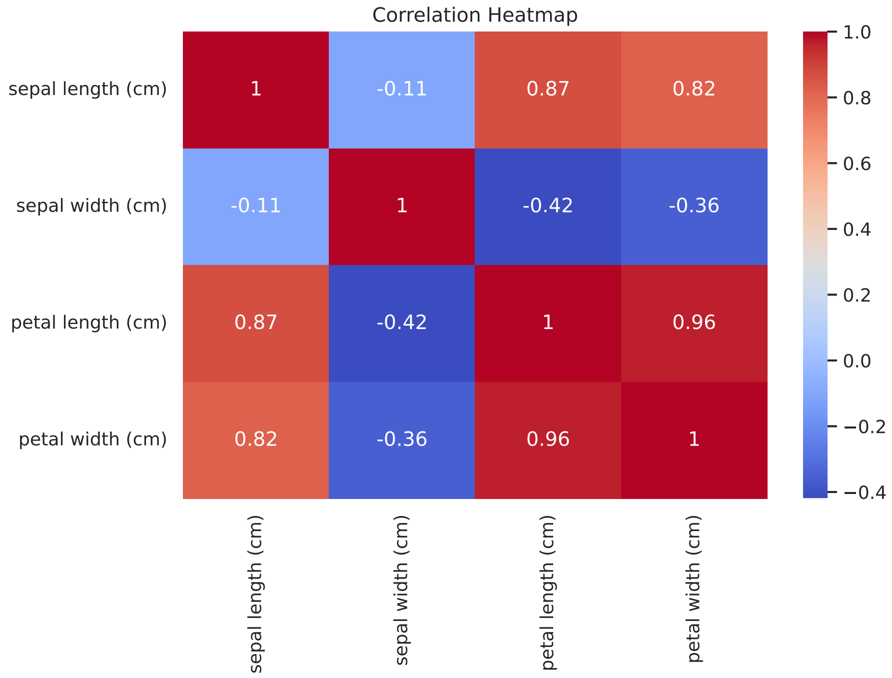

Figure 11: Pearson correlation matrix of the four numeric features

Inference: Petal length and petal width share the highest correlation (0.96), indicating strong co-variation. Both are strongly correlated with sepal length as well. Sepal width is weakly or negatively correlated with all other features, which explains its low Chi-Square and ANOVA scores. The high petal-to-petal correlation is why KNN performs well with just these two features — they pack most of the class-separating information into a tight 2D space.

2. Class Distribution

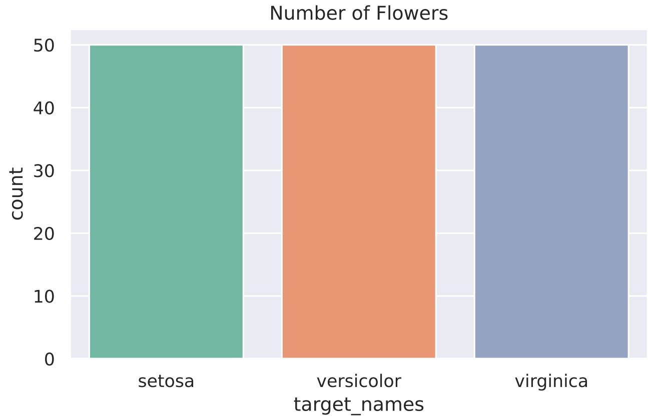

Figure 12: Number of samples per species (class distribution)

Inference: Each of the three species contributes exactly 50 samples, giving a perfectly balanced dataset. No class imbalance adjustments (such as oversampling or class-weighted loss) are necessary. Accuracy is therefore a reliable summary metric, and the 80/20 stratified split preserves this 1:1:1 ratio in both training and test sets.

3. Confusion Matrix

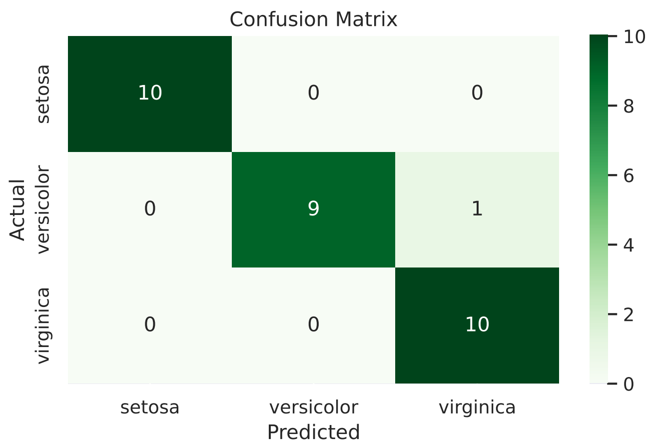

Figure 13: Confusion matrix of KNN predictions on the 30-sample test set

Inference: The model correctly predicted all 10 setosa and all 10 virginica samples. The single error is one versicolor sample classified as virginica, which is consistent with the slight petal-space overlap between these two species visible in the scatter plot. The off-diagonal cell (row versicolor, col virginica) equals 1, giving an overall test accuracy of $29/30 = 0.97$. Setosa remains perfectly separable from both other classes.

4. ROC Curves

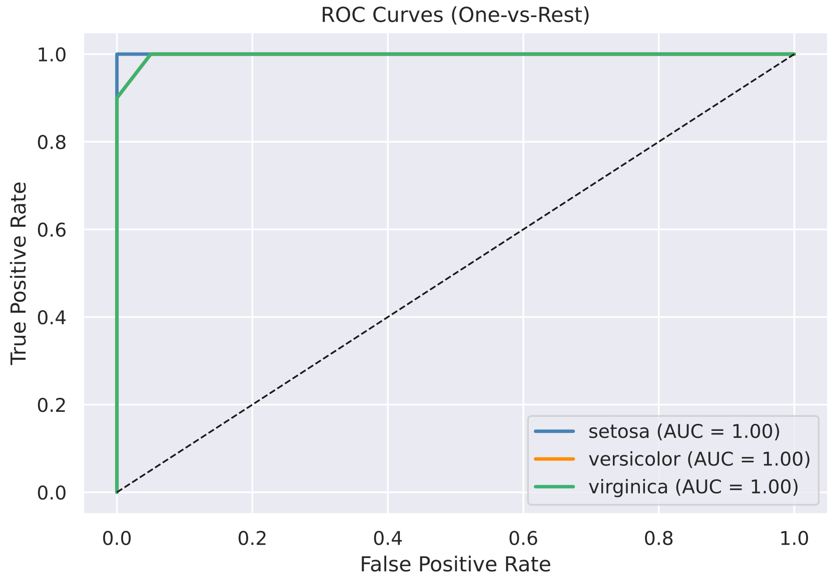

Figure 14: One-vs-Rest ROC curves for all three species

Inference: All three classes achieve an AUC of 1.00, meaning the probability scores produced by KNN perfectly rank positive samples above negative ones for every class. Setosa reaches the top-left corner immediately (no false positives at any threshold), while versicolor and virginica also trace the ideal curve despite the one borderline sample. An AUC of 1.00 for all classes on a 30-sample test set confirms strong generalisation of the learned 2D petal boundary.

5. Precision-Recall Curves

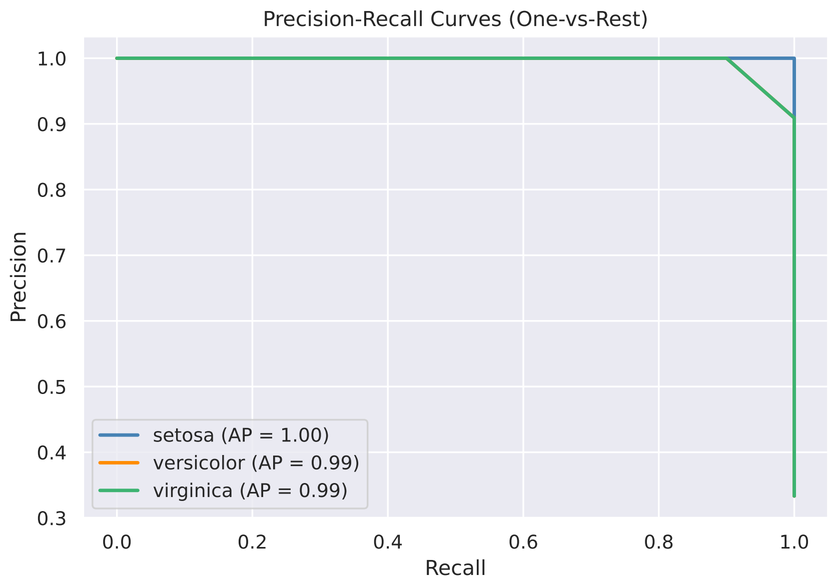

Figure 15: One-vs-Rest Precision-Recall curves for all three species

Inference: Setosa achieves perfect precision at all recall levels (AP = 1.00) since no other class encroaches on its petal-space region. Versicolor and virginica both obtain AP = 0.99, with a very small dip in precision at the highest recall threshold — corresponding to the single misclassified sample. The PR curves confirm the model is highly reliable even when class thresholds are varied, which is important for imbalanced deployment scenarios.

6. Learning Curve

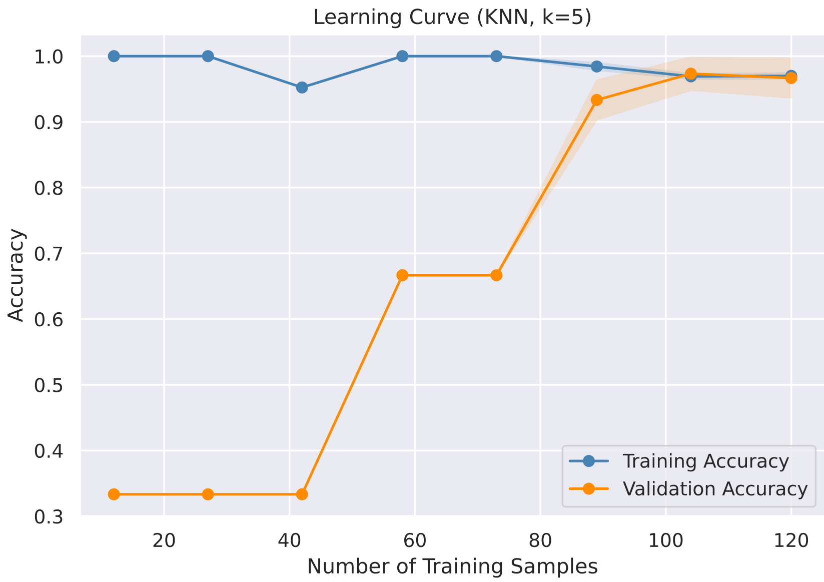

Figure 16: Training vs. validation accuracy as a function of training set size (KNN, $k = 5$, 5-fold CV)

Inference: Validation accuracy rises quickly and stabilises above 0.95 once around 60–80 training samples are available, indicating the model is not data-starved. The training accuracy stays at 1.0 throughout, which is typical of KNN ($k = 5$) on clean data — it memorises the training set perfectly. The narrow gap between training and validation accuracy at full data size confirms good generalisation with no overfitting. Adding more data beyond the current 120 training samples is unlikely to yield significant improvement.

7. Feature Importance (ANOVA Scores)

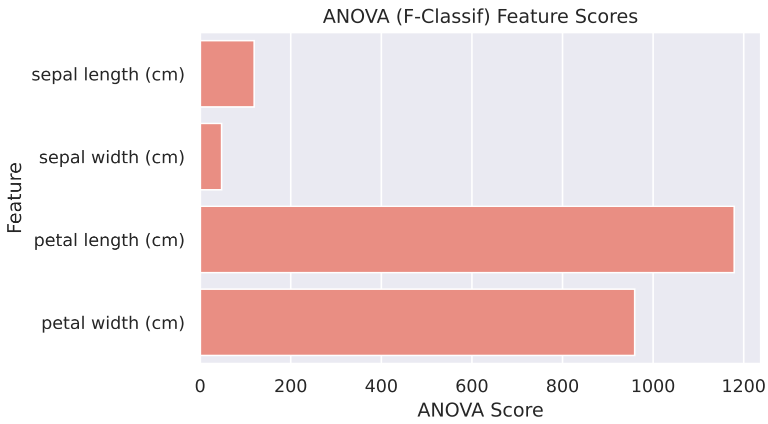

Figure 17: ANOVA F-scores for all four features (higher = more discriminative)

Inference: Petal length dominates with an F-score of 1179, nearly 10× higher than sepal length (119) and over 25× higher than sepal width (47). Petal width follows closely at 959. The enormous gap between petal and sepal scores quantitatively confirms what the pair plot shows visually: petal measurements are overwhelmingly more useful for separating the three species. Selecting only petal length and petal width therefore retains essentially all discriminative information while halving the feature space.

11 Discussion

The KNN classifier performed well on the Iris dataset, achieving 97% test accuracy using only two features (petal length and petal width) selected by ANOVA. Several points are worth noting:

- **Feature selection was effective.** Dropping sepal length and sepal width — the two lower-scoring features — did not hurt performance. This is consistent with the pair plot, which showed sepal features having more class overlap than petal features.
- **The single misclassification is expected.** The one versicolor sample predicted as virginica lies near the boundary of these two species in petal space. This is a known characteristic of the Iris dataset; some samples genuinely sit at the class boundary.
- **Standardisation mattered.** Without `StandardScaler`, features with larger absolute ranges (e.g., sepal length 4–8 cm) would dominate the Euclidean distance and unfairly suppress features with smaller ranges (e.g., petal width 0.1–2.5 cm).
- **Default $k = 5$ was sufficient.** Because the dataset is small and balanced, the default hyperparameters generalise well without tuning. Hyperparameter optimisation via GridSearchCV would be recommended for noisier or larger datasets.

- **Scalability:** KNN is not ideal for large-scale deployment since prediction time grows linearly with the number of training samples. For the 150-sample Iris dataset this is not an issue, but it would be a concern at scale.

12 Conclusion

This experiment demonstrated a complete supervised classification pipeline on the Iris dataset using K-Nearest Neighbours. EDA confirmed that petal length and petal width are the most discriminative features, and this was validated quantitatively by both Chi-Square and ANOVA feature selection tests. After standardisation and dimensionality reduction to two features, the default KNN classifier ($k = 5$) achieved 97% test accuracy with perfect classification of setosa and only one error on the versicolor-virginica boundary. The experiment confirms that KNN is a straightforward and effective baseline for well-structured, low-dimensional classification tasks, and that statistical feature selection can simplify the model without sacrificing accuracy.

13 References

1. Chitraju Vishnu Vineeth, "ML Lab Assignments — Assignment 1: Iris Classification," GitHub repository, 2026. [Online]. Available: https://github.com/VishnuVineeth14/ML_LAB_ASSIGNMENTS/tree/main/Assignment1